I Can turn an array to multiply

Level 3
Operations
Teacher 1

CCSS 3.OA.B.5 · **EE** none · access point M.EE.N.CN.2 · **CRA:** Concrete → Pictorial · **Flow:** Standard (3 new routine steps)

Lesson guidance:

Tip: Tray first, then the picture. Turn a real tray of counters before any drawn array.

Tip: No square in the I Do. A square turns into itself. It waits for the last row.

Watch: Recounting. A pupil who recounts the turned array has not got step 5 yet.

Lesson flow:

#	Section	Pupil page	Min
1	Prerequisite Check	Prerequisite Check	6
2	Number sense	Number sense	5
3	Spaced retrieval	Mini whiteboards	4
4	Chaining bridge	Mini whiteboards	2
5	I Do: the anchor chart	Anchor chart	10
6	We Do: Guided Practice	Lesson sheet	8
7	Check Question	Mini whiteboards	3
8	You Do: Independent Practice	Practice 1–2	9
9	Word problem	Word problem	5
10	Plenary and exit slip	Exit slip	4
	Transitions	between pages	4
	Total		**60**

Materials:

- ☐ 12 counters on a tray or card, one per pair
- ☐ Dot paper for the board
- ☐ Pupil packet, one per pupil
- ☐ Mini whiteboards and pens
- ☐ Anchor chart
- ☐ Exit slips, cut in half

Vocabulary:

• • • / • • •	• • • •	2 × 3 ↻ 3 × 2
array	**row**	**turn**
dots in equal rows	a line of dots across	move it a quarter way round

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I Can turn an array to multiply

Level 3
Operations
Teacher 2

What students need to know:

#	Skill or Concept	Atom Type	Teacher Notes
1	Count by 5	Fact · Pre-requisite	Prerequisite Check a.
2	Recall × 2 facts	Fact · Pre-requisite	Prerequisite Check b–e.
3	**Array fact:** rows × in each row	Conceptual · Pre-requisite	Prerequisite Check f–g.
4	**Turned array:** rows become columns	Comparative	Non-example: a dot added.
5	**Same product: turning adds no dot**	**Conceptual**	**Critical Atom. Ask "Did a dot move away?"**
6	Known fact gives the turned fact	Conceptual	Use the fact. Do not recount.

What students do:

Step	Task / Routine atom	Tag	Check it with
1	Look at the array: count the rows and in each row.	Pre-requisite	"How many rows?"
2	Write the fact.	Pre-requisite	Guided cell 1
3	Turn the array: rows become columns.	New	Tray: "How many rows now?"
4	Write the new fact.	New	Guided cell 2: three boxes
5	**Check: same product. No recount.**	**New**	**Check Question; exit b, c**

Rows tagged New: 3 → Flow: Standard. One chunk: turn the array; the product stays the same.

How to teach:

Turn a real tray of counters first: pupils count once, then see the same total. Ask "Did a dot move away?" First examples: non-square, no repeated digit.

Board: Dot paper: 3 rows of 2, then turned.

Common mistakes:

Mistake	Step	Looks like	Fix	Caught by
A new product	5	6 × 3 = 15	"Did a dot move away?"	Check Question A, Exit b
Adds the two factors	5	6 × 3 = 9	"Times, not plus: same product."	Check Question B
Copies the fact: no swap	4	9, 3, 27 for 3 × 9 = 27	Count the turned rows aloud.	Exit a · Practice 1
Product as a factor	4	45 in the box of exit c	The product is always last.	Exit c · Practice 2

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I Can turn an array to multiply

Level 3
Operations
Reference

Reference key:

Pages and order

Page	Tab	Score
P0	Prerequisite	one per question
P1	Number sense	/10
P2	Anchor chart	none (a poster)
P3	Lesson 1	Independent cells
P4	Practice n	every cell
P5	Word problem	/1
P6	Exit slip	/3 a half
T1–T5	Teacher n	—
T6	Answer Key	—

Fade levels

Label	Means
★	Guided: hints on, step 1 traced
★★	Independent: hints off
★★★	Last row: the edge cases

Data loop

● **Most:**	Go on.
◑ **About half:**	Short reteach, then go on.
○ **Fewer than half:**	Stop. Reteach with a concrete example.

Abbreviations

Short	In full
MWBs	mini whiteboards
ELL	English language learners
SPED	special education students
RCDWS	Read, Circle, Draw, Write, Solve
CRA	Concrete-Representational-Abstract

Routing ids

No site lesson exists yet for these prerequisites: T6 names a CCSS code and a short title (V22 Appendix H).

Answer slots

Shape	Used for
_____	a number after =
☐	a part inside an expression
☐	a tick (routine panel, exit slip)

Icons

Icon	Means
	Tip
	Feedback
	Watch
	Research
	Critical Atom
	Language tip
	Why it works
	Board
1 2 3 4 5 ☑	The pupil Steps (accent)

Band labels

Pupil: Vocabulary · Remember · Guided Practice · Steps · Independent Practice · Mixed Review · Number sense: A / B · Workspace · Another example · Say · Rule.

Teacher: Lesson guidance · Lesson flow · Materials · What students need to know · What students do · How to teach · Common mistakes · Section n · Prerequisite routing · Accepted answers · Answers at a glance.

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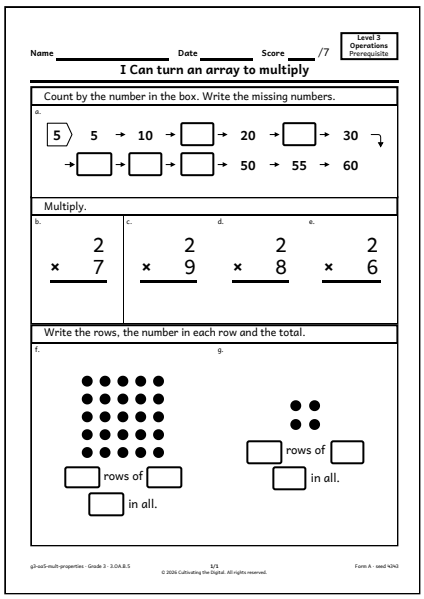
I Can turn an array to multiply

Level 3 Operations Teacher 3

Section 1 · Prerequisite Check (6 min)

Tip: Route before teaching. Mark it while pupils do the Number sense page.

1. **Do:** Hand out the Prerequisite Check, face down.
2. **Say:** "Do these on your own. Try every one."
3. **Do:** Walk and mark as pupils finish.
   Note: Expected: **a. 15, 25, 35, 40, 45 · b. 14 · c. 18 · d. 16 · e. 12 · f. 25 · g. 4**. Routing is on the last page (T6).



Name _____ Date _____ Score ___ /7 — Level 3 Operations Prerequisite

I Can turn an array to multiply

Count by the number in the box. Write the missing numbers.

a. 5 → 5 → 10 → ☐ → 20 → ☐ → 30 → ☐ → ☐ → ☐ → 50 → 55 → 60

Multiply.

b. 2 × 7 c. 2 × 9 d. 2 × 8 e. 2 × 6

Write the rows, the number in each row and the total.

f. ☐ rows of ☐ ☐ in all.

g. ☐ rows of ☐ ☐ in all.

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Section 2 · Number sense (5 min)

1. **Do:** Hand out the Number sense page.
2. **Say:** "Band A: read each row. Write the answer."
3. **Ask:** "What is 10 more than 47?"
   Note: Expected: **57**. Band B: the same five kinds, harder numbers.

Section 3 · Spaced retrieval (4 min)

1. **Do:** Pupils hold mini whiteboards. Write each fact on the board.
2. **Ask:** "5 × 8 = ?"
   Note: Expected: **40**.
3. **Ask:** "2 × 10 = ?"
   Note: Expected: **20**.
4. **Ask:** "10 × 5 = ?"
   Note: Expected: **50**.

Section 4 · Chaining bridge (2 min)

1. **Say:** "Last time: one array gives a fact."
2. **Say:** "Today: one array gives two facts. Same answer."
3. **Do:** Show the tray. Pupils count the counters once.
   Note: They keep the total in mind for the turn.

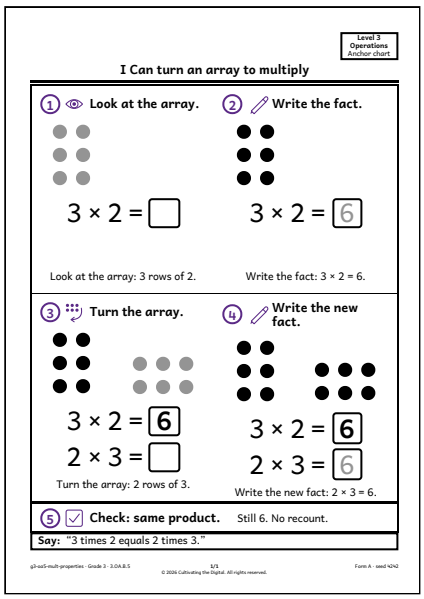
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I Can turn an array to multiply

Level 3 Operations Teacher 4

Section 5 · I Do: the anchor chart (10 min)

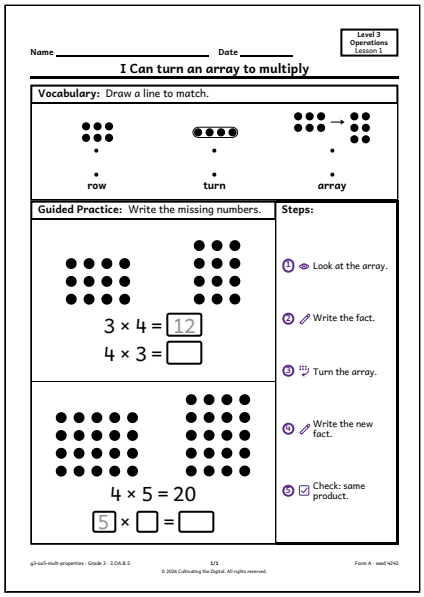
Watch: Turn slowly. Let them see that no dot leaves the tray.



1. **Do:** Hold up a tray of 3 rows of 2.
2. **Say:** "3 rows of 2 is 6."
   Note: Step 1 and 2: write **3 × 2 = 6**.
3. **Do:** Turn the tray a quarter way round. Slowly.
   Note: Step 3.
4. **Ask:** "How many rows now? How many in each row?"
   Note: Expected: **2** rows of **3**.
5. **Say:** "2 rows of 3 is 6."
   Note: Step 4: write **2 × 3 = 6**.
6. **Ask:** "Did a dot move away?"
   Note: Expected: no. Step 5: same product.
7. **Say:** "3 times 2 equals 2 times 3."
   Note: The Say frame. Pupils repeat twice.
8. **Ask:** "Cover the dots. So what is 2 × 3?"
   Note: Expected: **6**. No dots: same two numbers, same product.
9. **Do:** Write 2 × □ = 6 on the board.
   Note: Expected: **3**: the factor that moved.
10. **Do:** Write 3 × 2 = 6, then three boxes under it. Point to the turned tray.
    Note: Fill them: **2** × **3** = **6**. Guided cell 2.

Section 6 · We Do: Guided Practice (8 min)

Feedback: Fix it at once. Fix: turn the tray again and ask "Same dots?"



1. **Say:** "Look at the first array. Count the rows."
2. **Ask:** "How many rows? How many in each row?"
   Note: Expected ★: **3 × 4 = 12**; turned **4 × 3 = 12**.
3. **Say:** "Now the fact is given. Write the new fact."
   Note: The first box is traced: the rows of the turned array.
4. **Do:** Pupils do the second array on their own.
   Note: Expected ★: **5 × 4 = 20**, from **4 × 5 = 20**.
5. **Ask:** "Same product? Did you count again?"
   Note: Expected: same product, no recount. Step 5.

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I Can turn an array to multiply

Level 3 Operations Teacher 5

Section 7 · Check Question (3 min)

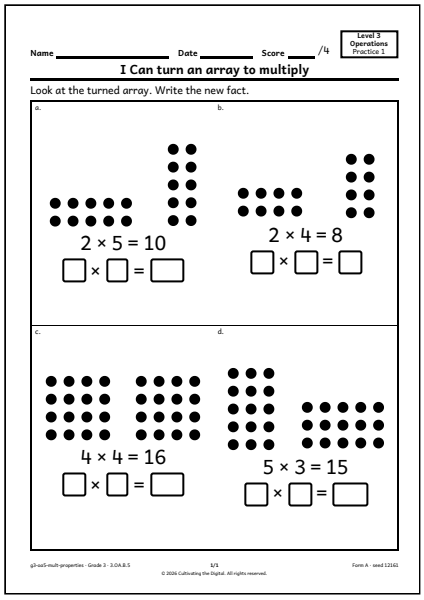
1. **Do:** Write 3 × 6 = 18 on the board. No dots.
2. **Do:** Write three turned facts. A: 6 × 3 = 15 · B: 6 × 3 = 9 · C: 6 × 3 = 18.
3. **Ask:** "Which turned fact is true? Show fingers."
   Note: Correct: **C 6 × 3 = 18**. A: A new product · B: Adds the two factors

- ● **Most:** Go on to You Do.
- ◑ **About half:** Reteach step 5 with the tray, then two more.
- ○ **Fewer than half:** Stop. Whole class: We Do again with a new tray.

Section 8 · You Do: Independent Practice (9 min)

Watch: The last row. Walk to the last row first: Practice 1: 16 (a square), 15 (a 5); Practice 2: 49 (a square), 35 (a 5).

1. **Do:** Pupils finish the lesson sheet, then Practice 1 and 2.
2. **Say:** "Look at the turned array. Write the new fact. Check: same product."
3. **Do:** Circulate at ★★. Look for pupils who recount.
   Note: Ask "Did a dot move away?"


Name _____ Date _____ Score ___/4 Level 3 Operations Practice 1

I Can turn an array to multiply

Look at the turned array. Write the new fact.

a. 2 × 5 = 10 □ × □ = □

b. 2 × 4 = 8 □ × □ = □

c. 4 × 4 = 16 □ × □ = □

d. 5 × 3 = 15 □ × □ = □

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Section 9 · Word problem (5 min)

1. **Do:** Read the story aloud two times. Pupils tick each box.
2. **Ask:** "How many rows now? How many cakes in each row?"
   Note: Expected: **6** rows, **5** in each.
3. **Ask:** "Say both facts."
   Note: Expected: 5 × 6 = 30 and 6 × 5 = 30.

Section 10 · Plenary and exit slip (4 min)

1. **Say:** "Turning does not change the total."
   Note: Say it together twice.
2. **Do:** Hand out the exit slip. No help. Then sort the slips.
   Note: a. **3, 9, 27** · b. **42** · c. **9**.

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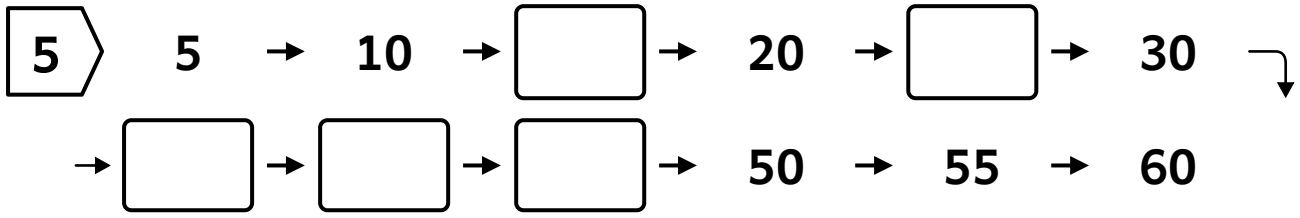
Name _____ Date _____ Score _____ /7

Level 3
Operations
Prerequisite

I Can turn an array to multiply

Count by the number in the box. Write the missing numbers.

a.



Multiply.

b.

$$\begin{array}{r} 2 \\ \times\ 7 \\ \hline \end{array}$$

c.

$$\begin{array}{r} 2 \\ \times\ 9 \\ \hline \end{array}$$

d.

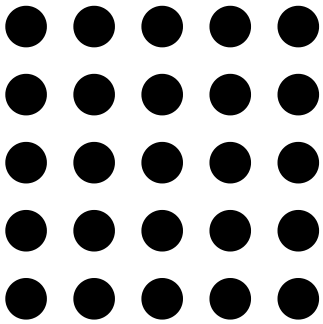
$$\begin{array}{r} 2 \\ \times\ 8 \\ \hline \end{array}$$

e.

$$\begin{array}{r} 2 \\ \times\ 6 \\ \hline \end{array}$$

Write the rows, the number in each row and the total.

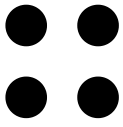
f.



_____ rows of _____

_____ in all.

g.



_____ rows of _____

_____ in all.

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Name _____ Date _____ Score _____ /10

Level 3
Operations
Number sense

I Can turn an array to multiply

Number sense: A

1 Write the missing number.

10 more than **47** is _____ .

2 Write the missing number.

10 less than **20** is _____ .

3 Write <, > or = in the circle.

72 ◯ **35**

4 Write how many of each place.

39 = ☐ tens ☐ ones

5 Write the missing number.

1 more than **33** is _____ .

Number sense: B

6 Write the missing number.

10 more than **877** is _____ .

7 Write the missing number.

10 less than **305** is _____ .

8 Write <, > or = in the circle.

544 ◯ **565**

9 Write how many of each place.

84 = ☐ tens ☐ ones

10 Write the missing number.

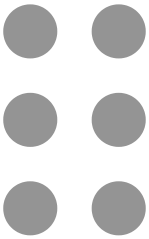
1 more than **103** is _____ .

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Level 3
Operations
Anchor chart

I Can turn an array to multiply

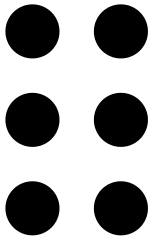
1 Look at the array.



$3 \times 2 = \square$

Look at the array: 3 rows of 2.

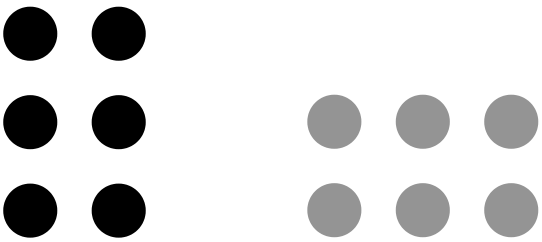
2 Write the fact.



$3 \times 2 = 6$

Write the fact: 3 × 2 = 6.

3 Turn the array.

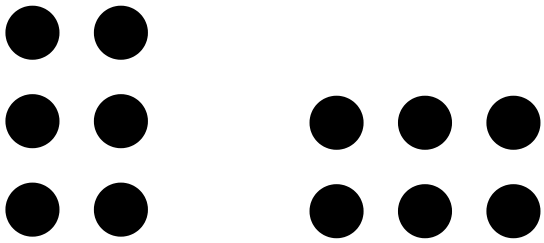


$3 \times 2 = 6$

$2 \times 3 = \square$

Turn the array: 2 rows of 3.

4 Write the new fact.



$3 \times 2 = 6$

$2 \times 3 = 6$

Write the new fact: 2 × 3 = 6.

5 Check: same product.

Still 6. No recount.

Say: "3 times 2 equals 2 times 3."

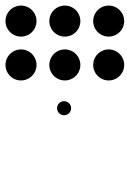
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Name _____ Date _____

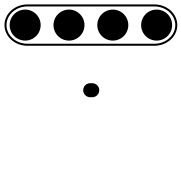
Level 3
Operations
Lesson 1

I Can turn an array to multiply

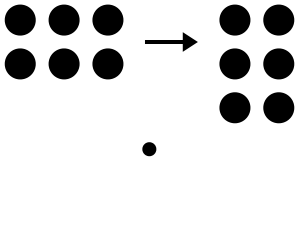
Vocabulary: Draw a line to match.



row

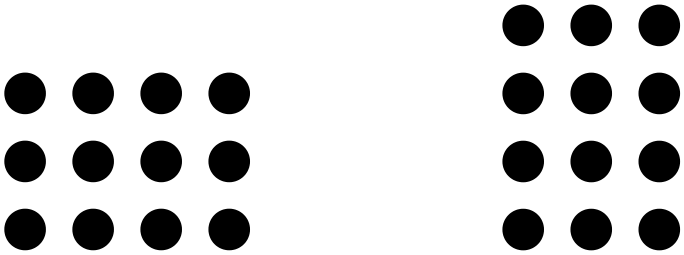


turn



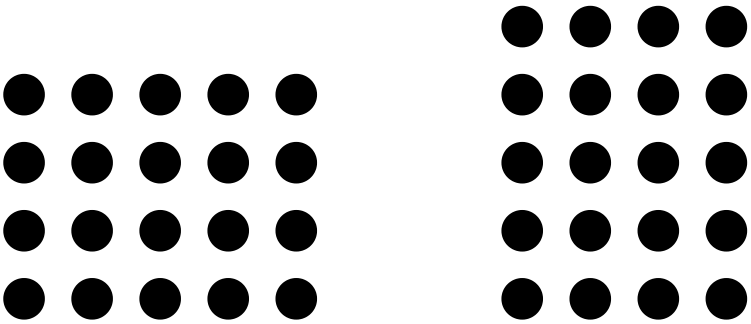
array

Guided Practice: Write the missing numbers.



$3 \times 4 =$ 12

$4 \times 3 =$ ____



$4 \times 5 = 20$

5 × ____ = ____

Steps:

1. Look at the array.
2. Write the fact.
3. Turn the array.
4. Write the new fact.
5. Check: same product.

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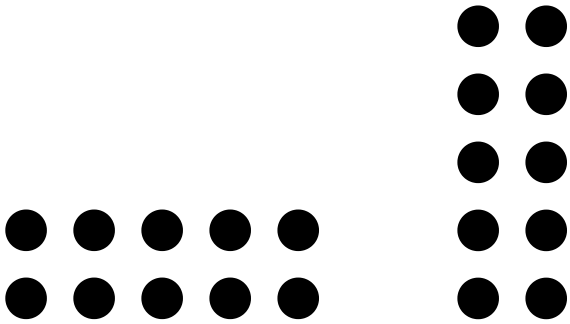
Name _____ Date _____ Score _____ /4

Level 3 Operations Practice 1

I Can turn an array to multiply

Look at the turned array. Write the new fact.

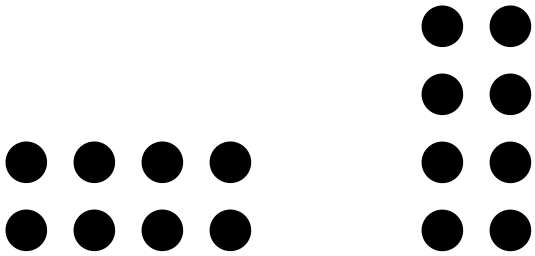
a.



$2 \times 5 = 10$

$\square \times \square = \square$

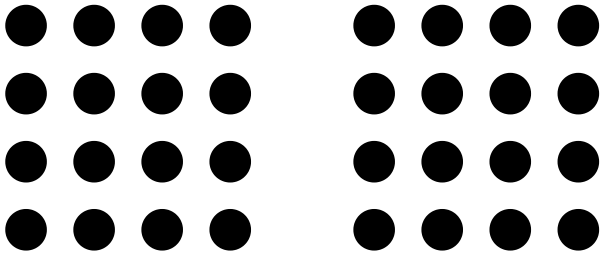
b.



$2 \times 4 = 8$

$\square \times \square = \square$

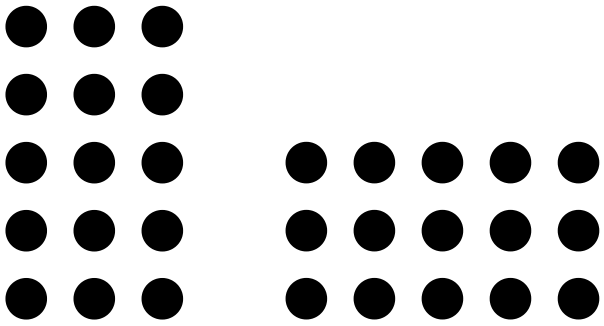
c.



$4 \times 4 = 16$

$\square \times \square = \square$

d.



$5 \times 3 = 15$

$\square \times \square = \square$

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Name _____ Date _____ Score _____ /10

Level 3
Operations
Practice 2

I Can turn an array to multiply

Write the missing number.

a. $7 \times 9 = 63$ $9 \times 7 = \square$	b. $8 \times 3 = 24$ $3 \times 8 = \square$
c. $4 \times 7 = 28$ $7 \times \square = 28$	d. $4 \times 9 = 36$ $9 \times 4 = \square$
e. $9 \times 6 = 54$ $6 \times 9 = \square$	f. $8 \times 4 = 32$ $4 \times 8 = \square$
g. $8 \times 9 = 72$ $9 \times \square = 72$	h. $6 \times 8 = 48$ $8 \times 6 = \square$
i. $7 \times 7 = 49$ $7 \times 7 = \square$	j. $7 \times 5 = 35$ $5 \times 7 = \square$

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Name _____ Date _____ Score _____ /1

Level 3
Operations
Word problem

I Can turn an array to multiply

Read the story. Draw it. Solve.

A tray has 5 rows of 6 cakes.

Sam turns it: now it has 6 rows.

How many in each row now?

There are _____ cakes in each row.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

Workspace:

☐ × ☐ = ☐

☐ × ☐ = ☐

Say: "5 times 6 equals 6 times _____."

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Name _____ Score _____ /3

Level 3 Operations Exit slip

I Can turn an array to multiply

Turn the fact. Write the new fact.

a.

$$9 \times 3 = 27$$

$$\square \times \square = \square$$

Write the missing number.

b.

$$6 \times 7 = 42$$

$$7 \times 6 = \square$$

c.

$$9 \times 5 = 45$$

$$5 \times \square = 45$$

Say: "Same two numbers, same product."

☐ I can do this. ☐ I need one more day.

Name _____ Score _____ /3

Level 3 Operations Exit slip

I Can turn an array to multiply

Turn the fact. Write the new fact.

a.

$$9 \times 3 = 27$$

$$\square \times \square = \square$$

Write the missing number.

b.

$$6 \times 7 = 42$$

$$7 \times 6 = \square$$

c.

$$9 \times 5 = 45$$

$$5 \times \square = 45$$

Say: "Same two numbers, same product."

☐ I can do this. ☐ I need one more day.

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Answer Key _____ **Date** _____ **Score** _____ /7

Level 3 Operations Answer Key

I Can turn an array to multiply

Count by the number in the box. Write the missing numbers.

a.

5 → 5 → 10 → 15 → 20 → 25 → 30 →

→ 35 → 40 → 45 → 50 → 55 → 60

Multiply.

b.

$$\begin{array}{r} 2 \\ \times\ 7 \\ \hline 14 \end{array}$$

c.

$$\begin{array}{r} 2 \\ \times\ 9 \\ \hline 18 \end{array}$$

d.

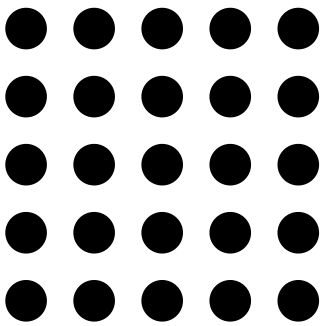
$$\begin{array}{r} 2 \\ \times\ 8 \\ \hline 16 \end{array}$$

e.

$$\begin{array}{r} 2 \\ \times\ 6 \\ \hline 12 \end{array}$$

Write the rows, the number in each row and the total.

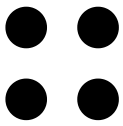
f.



5 rows of 5

25 in all.

g.



2 rows of 2

4 in all.

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Answer Key _____ Date _____ Score _____ /10

Level 3 Operations Answer Key

I Can turn an array to multiply

Number sense: A

1. Write the missing number. **10** more than **47** is **57**.
2. Write the missing number. **10** less than **20** is **10**.
3. Write <, > or = in the circle. **72** **>** **35**
4. Write how many of each place. **39** = **3** tens **9** ones
5. Write the missing number. **1** more than **33** is **34**.

Number sense: B

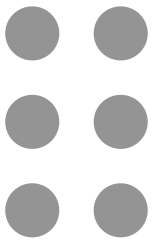
6. Write the missing number. **10** more than **877** is **887**.
7. Write the missing number. **10** less than **305** is **295**.
8. Write <, > or = in the circle. **544** **<** **565**
9. Write how many of each place. **84** = **8** tens **4** ones
10. Write the missing number. **1** more than **103** is **104**.

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Level 3
Operations
Answer Key

I Can turn an array to multiply

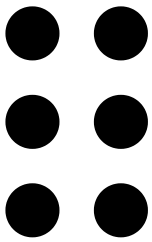
1 Look at the array.



$3 \times 2 = \square$

Look at the array: 3 rows of 2.

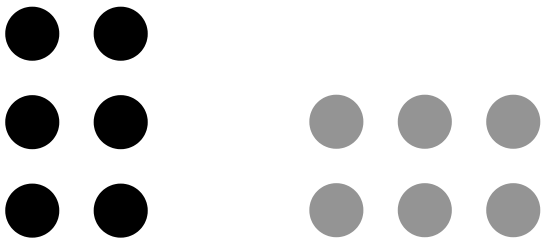
2 Write the fact.



$3 \times 2 = 6$

Write the fact: 3 × 2 = 6.

3 Turn the array.

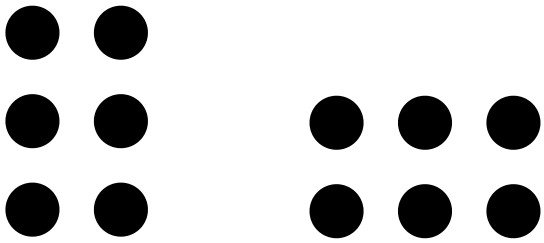


$3 \times 2 = 6$

$2 \times 3 = \square$

Turn the array: 2 rows of 3.

4 Write the new fact.



$3 \times 2 = 6$

$2 \times 3 = 6$

Write the new fact: 2 × 3 = 6.

5 Check: same product.

Still 6. No recount.

Say: "3 times 2 equals 2 times 3."

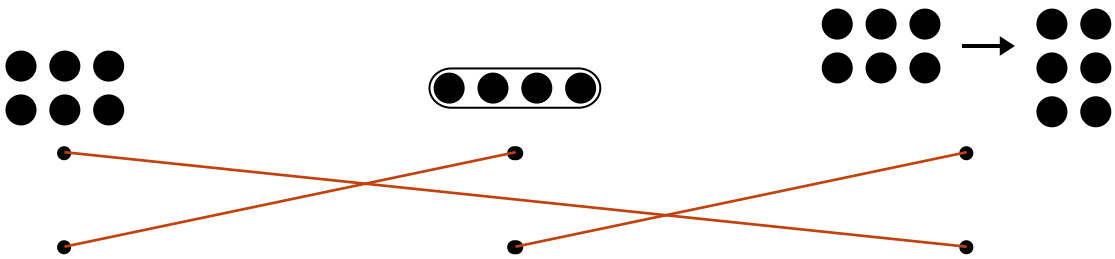
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Answer Key _____ Date _____

Level 3 Operations Answer Key

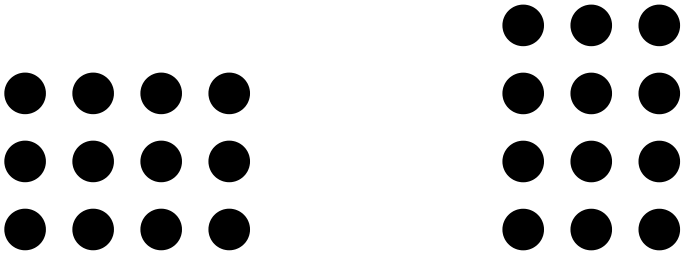
I Can turn an array to multiply

Vocabulary: Draw a line to match.



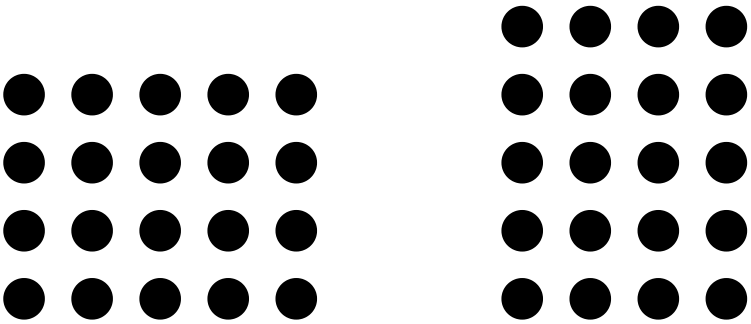
row turn array

Guided Practice: Write the missing numbers.



$3 \times 4 = \boxed{12}$

$4 \times 3 = \boxed{12}$



$4 \times 5 = 20$

$\boxed{5} \times \boxed{4} = \boxed{20}$

Steps:

1. Look at the array.
2. Write the fact.
3. Turn the array.
4. Write the new fact.
5. Check: same product.

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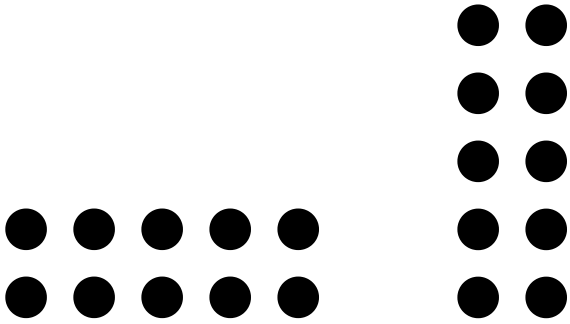
Answer Key _____ **Date** _____ **Score** _____ /4

Level 3 Operations Answer Key

I Can turn an array to multiply

Look at the turned array. Write the new fact.

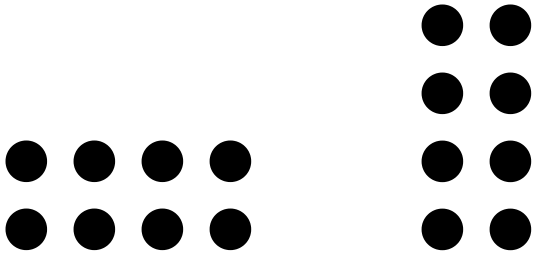
a.



$2 \times 5 = 10$

$\boxed{5} \times \boxed{2} = \boxed{10}$

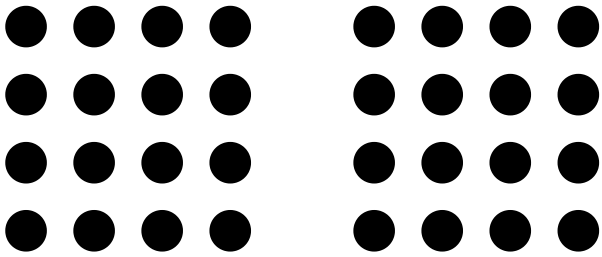
b.



$2 \times 4 = 8$

$\boxed{4} \times \boxed{2} = \boxed{8}$

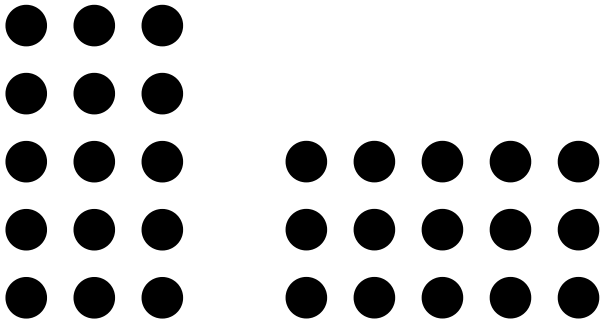
c.



$4 \times 4 = 16$

$\boxed{4} \times \boxed{4} = \boxed{16}$

d.



$5 \times 3 = 15$

$\boxed{3} \times \boxed{5} = \boxed{15}$

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Answer Key _____ **Date** _____ **Score** _____ /10

Level 3 Operations Answer Key

I Can turn an array to multiply

Write the missing number.

a.

$7 \times 9 = 63$

$9 \times 7 =$ **63**

b.

$8 \times 3 = 24$

$3 \times 8 =$ **24**

c.

$4 \times 7 = 28$

$7 \times$ **4** $= 28$

d.

$4 \times 9 = 36$

$9 \times 4 =$ **36**

e.

$9 \times 6 = 54$

$6 \times 9 =$ **54**

f.

$8 \times 4 = 32$

$4 \times 8 =$ **32**

g.

$8 \times 9 = 72$

$9 \times$ **8** $= 72$

h.

$6 \times 8 = 48$

$8 \times 6 =$ **48**

i.

$7 \times 7 = 49$

$7 \times 7 =$ **49**

j.

$7 \times 5 = 35$

$5 \times 7 =$ **35**

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Answer Key _____ **Date** _____ **Score** _____ /1

Level 3 Operations Answer Key

I Can turn an array to multiply

Read the story. Draw it. Solve.

A tray has 5 rows of 6 cakes.

Sam turns it: now it has 6 rows.

How many in each row now?

There are **5** cakes in each row.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

Workspace:

5 × **6** = **30**

6 × **5** = **30**

Say: "5 times 6 equals 6 times **5**."

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Answer Key _____ **Score** _____ /3

Level 3 Operations Exit slip

I Can turn an array to multiply

Turn the fact. Write the new fact.

a.

$9 \times 3 = 27$

$\boxed{3} \times \boxed{9} = \boxed{27}$

Write the missing number.

b.

$6 \times 7 = 42$

$7 \times 6 = \boxed{42}$

c.

$9 \times 5 = 45$

$5 \times \boxed{9} = 45$

Say: "Same two numbers, same product."

☐ I can do this. ☐ I need one more day.

Answer Key _____ **Score** _____ /3

Level 3 Operations Exit slip

I Can turn an array to multiply

Turn the fact. Write the new fact.

a.

$9 \times 3 = 27$

$\boxed{3} \times \boxed{9} = \boxed{27}$

Write the missing number.

b.

$6 \times 7 = 42$

$7 \times 6 = \boxed{42}$

c.

$9 \times 5 = 45$

$5 \times \boxed{9} = 45$

Say: "Same two numbers, same product."

☐ I can do this. ☐ I need one more day.

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I Can turn an array to multiply

Level 3
Operations
Answer Key

Prerequisite routing:

Question	Prerequisite (T2 atom)	If missed → Lesson
a. Count by 5	Fact, atom 1	2.NBT.A.2 Count by 5s
b–e. Recall × 2 facts	Fact, atom 2	3.OA.C.7 Multiply by 2
f–g. Array fact	Conceptual, atom 3	2.OA.C.4 Use arrays
	Rule	0 missed: go on · 1: that lesson · 2+: the earliest

Accepted answers:

- Prerequisite Check arrays: rows and in each row in either order.
- Exit a: only 3 × 9 = 27; writing 9 × 3 = 27 again is the no-swap mistake.
- Practice 2 and exit b, c: one number in each box, no working needed.
- Practice 1: a turned fact read off the turned array is right even if the pupil counted it.
- Cake story: 5 in each row; the two facts 5 × 6 = 30 and 6 × 5 = 30, in either order; the label "cakes" may be left off.

Answers at a glance:

Page	Answers	Fade
P0 Prerequisite	**15, 25, 35, 40, 45 · 14 · 18 · 16 · 12 · 25 · 4**	★★
P1 Number sense	**57 · 10 · > · 3 tens 9 ones · 34 · 887 · 295 · < · 8 tens 4 ones · 104**	★★
P3 Guided	**12, 12 · 5, 4, 20**	★
P4-1 Practice, row 1	**5, 2, 10 · 4, 2, 8**	★★
P4-1 Practice, last row	**4, 4, 16 · 3, 5, 15**	★★★
P4-2 Practice, rows 1–4	**63 · 24 · 4 · 36 · 54 · 32 · 8 · 48**	★★
P4-2 Practice, last row	**49 · 35**	★★★
P5 Word problem	**5 × 6 = 30 · 6 × 5 = 30**	★
P6 Exit slip	**a. 3, 9, 27 · b. 42 · c. 9**	★★★

Wrong answers to look for when marking:

Pupil wrote	Likely mistake	Next step
36 (b)	**A new product**	Tray: turn it; "Did a dot move away?"
9, 3, 27 (a)	**Copies the fact: no swap**	Count the turned rows aloud.
45 (c)	**Product as a factor**	The box map: the product is last.
13 (b)	**Adds the two factors**	"Times, not plus." Say the fact aloud.
The first fact copied on Practice 1	**Copies the fact: no swap**	Turn the page a quarter way: count the rows.
A × 2 fact missed on the Prerequisite Check	**Not yet a × 2 fact**	3.OA.C.7: count by 2 to find it.
A box left empty on Practice 2	**Not sure which number moved**	Point: the two factors swap places.

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